Notes: Ederer & Pellegrino (RES, 2026)
A Tale of Two Networks: Common Ownership and Product Market Rivalry

Contents

1 Big picture

- **Research question.** How large are the economy-wide welfare and distributional effects of common ownership among U.S. public firms?
- **Two-network mechanism.** The paper studies the overlap between the product-rivalry network and the ownership-overlap network.
- **Main finding.** Under proportional influence, the welfare cost of common ownership rises about ninefold from 1995 to 2021.
- **2021 result.** Common ownership lowers total surplus relative to standard Cournot and transfers surplus from consumers to producers.

2 Data and measurement

Compustat provides revenues, costs, SG&A, and profits. Hoberg–Phillips text-based product similarities provide the product-rivalry network. SEC 13(f) filings and share data provide the ownership network. Table 1 maps all observed and identified variables into the model.

3 Models and empirical specifications

3.1 GHL demand

$$p(q) = b - (I + \Sigma)q.$$

3.2 Common ownership objective

$$\phi_i(q) = \pi_i(q) + \sum_j \kappa_{ij}\pi_j(q).$$

3.3 Cournot with common ownership

$$q^{\phi} = (2I + \Delta + \Sigma + K \circ \Sigma)^{-1}(b - c^0).$$

Detailed interpretation: The object $K \circ \Sigma$ captures the overlap of ownership and product-market rivalry.

4 Identification and empirical design

The paper is a structural quantification exercise. Identification comes from combining demand restrictions, ownership data, product similarity data, and firm financials. The comparison between common ownership and standard Cournot isolates the incremental product-market effect of common ownership.

5 Section-by-section study guide

- Section 2 develops the GHL demand and Cournot network model.
- Section 3 maps theory to data and validates the demand system.

- Section 4 presents baseline empirical welfare estimates.
- Section 5 studies alternative governance assumptions.
- Section 6 studies extensions and robustness.

6 Tables and figures

6.1 Table 1: Variable Definitions and Mapping to Data

Read: Table 1 maps observed variables and identified objects to data sources.

Detailed interpretation: This table is the bridge from the model to the empirical implementation.

Economic message: The welfare estimates depend on this mapping from firm data to structural primitives.

Table 1: Variable Definitions and Mapping to Data

Panel A: Observed Variables

Notation	Description	Measurement	Source
$p_i q_i$	Revenues	Revenues	Compustat
TVC_i	Total Variable Costs	Costs of Goods Sold	Compustat
f_i	Fixed Costs	Selling, General and Administrative Costs	Compustat
$\mathbf{a}_i'\mathbf{a}_j$	Product Cosine Similarity	Word frequencies in 10-K Business Description	Hoberg and Phillips (2016)
$\mathbf{s}_i$	Ownership	Number of shares divided by total shares outstanding	SEC 13(f) filings, Compustat

Panel B: Identified Variables and Parameters

Notation	Description	Identification
δ_i	Marginal Cost Slope	$= 0$ in baseline model
α	Utility Weight on Common Characteristics	$= 0.12$ using Nevo (2001) as in Pellegrino (2019)
q_i	Output	$\mathbf{q}$ such that $\boldsymbol{\pi} + \mathbf{f} = \mathrm{diag}(\mathbf{q})\left(\mathbf{I} + \frac{1}{2}\boldsymbol{\Delta} + \mathbf{K} \circ \boldsymbol{\Sigma}\right)\mathbf{q}$
c_i^0	Marginal Cost Intercept	$= \frac{h_i}{q_i} - \frac{1}{2}\delta_i q_i$
$(\mathbf{I} + \boldsymbol{\Sigma})$	$\partial \mathbf{p} / \partial \mathbf{q}$	$= (1-\alpha)\,\mathbf{I} + \alpha \mathbf{A}'\mathbf{A}$
$\mathbf{b}$	Demand Intercept	$= (2\mathbf{I} + \boldsymbol{\Delta} + \boldsymbol{\Sigma} + \mathbf{K} \circ \boldsymbol{\Sigma})\,\mathbf{q} + \mathbf{c}^0$

Table 2: Microeconometric Estimates vs. GHL (Pellegrino, 2019)

			Demand Elasticity $\left(\frac{\partial q_i}{\partial p_j} \cdot \frac{p_j}{q_i}\right)$	
Market	**Firm i**	**Firm j**	**Micro Estimate**	**GHL**
Auto	Ford	Ford	-4.320	-5.197
Auto	Ford	General Motors	0.034	0.056
Auto	Ford	Toyota	0.007	0.017
Auto	General Motors	Ford	0.065	0.052
Auto	General Motors	General Motors	-6.433	-4.685
Auto	General Motors	Toyota	0.008	0.005
Auto	Toyota	Ford	0.018	0.025
Auto	Toyota	General Motors	0.008	0.008
Auto	Toyota	Toyota	-3.085	-4.851
Cereals	Kellogg's	Kellogg's	-3.231	-1.770
Cereals	Kellogg's	Quaker Oats	0.033	0.023
Cereals	Quaker Oats	Kellogg's	0.046	0.031
Cereals	Quaker Oats	Quaker Oats	-3.031	-1.941
Computers	Apple	Apple	-11.979	-8.945
Computers	Apple	Dell	0.018	0.025
Computers	Dell	Apple	0.027	0.047
Computers	Dell	Dell	-5.570	-5.110

6.2 Table 2: Demand Elasticities: Microeconometric Estimates versus GHL

Read: The table compares GHL elasticities to classic IO estimates.

Detailed interpretation: Signs match across all shown firm pairs and magnitudes are broadly plausible.

Economic message: Product-similarity data generate credible demand elasticities.

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Table 2: Demand Elasticities

6.3 Figure 1: Network Visualisation

Read: Product similarity has clustered communities; common ownership has a hub-and-spoke structure.

Detailed interpretation: The two networks are visually distinct, which is why their overlap matters.

Economic message: Common ownership is a network problem.

6.4 Figure 2: Network Evolution–Product Similarity and Ownership

Read: Ownership becomes much more connected from 1995 to 2021, while product similarity changes less.

Detailed interpretation: This time trend explains why common-ownership distortions grow.

Economic message: The ownership network is the main changing force.

6.5 Figure 3 and Figure 4: Product Similarity, Profit Weights, and Combined Network

Read: Figure 3 shows the joint distribution of rivalry and ownership weights. Figure 4 overlays the two networks.

Figure 1: Network Visualization

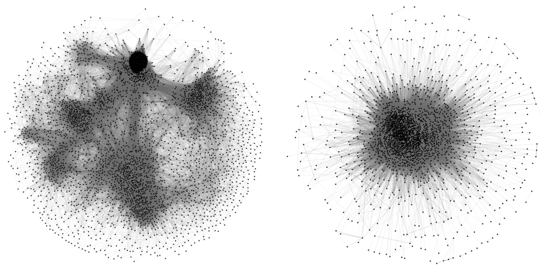
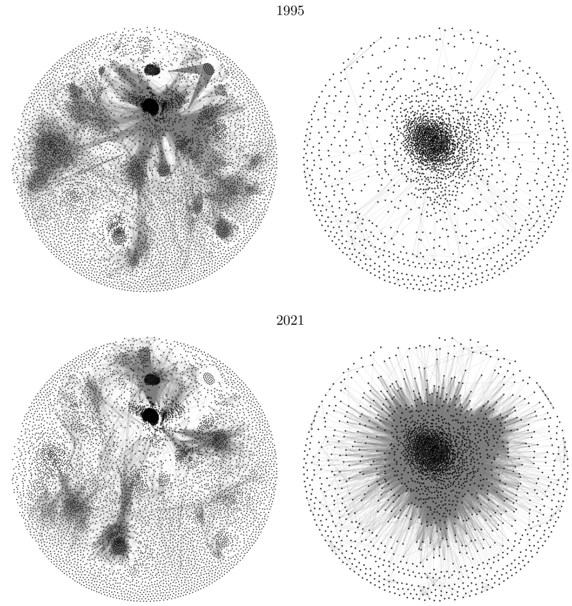


Figure Notes: The diagrams are two-dimensional representations of the network of product similarities (left panel), first computed by Hoberg and Phillips (2016), and of the network of ownership shares (right panel). The data include the universe of publicly listed firms in 2008. Firm pairs that are closer in product market space and closer in ownership space, are shown with thicker links. These distances are computed in spaces that have approximately 61,000 and 2,800 dimensions, respectively. We use the OpenOrd algorithm of Martin et al. (2011) and the FR gravity algorithm of Fruchterman and Reingold (1991) to plot these high-dimensional objects over a plane.

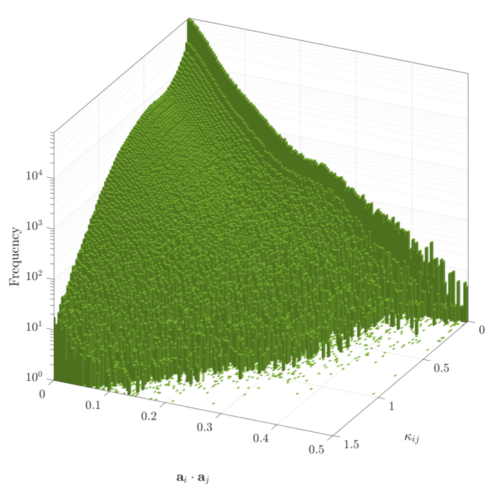
Figure 2: Network Evolution - Product Similarity and Ownership



Detailed interpretation: Most firm pairs do not overlap strongly in both dimensions, but the pairs that do are central for welfare.

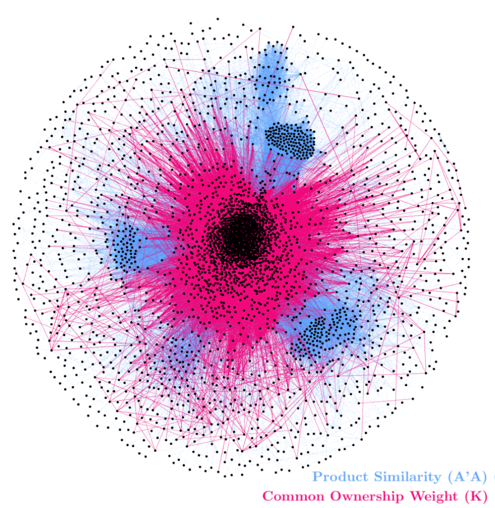
Economic message: The distortion is governed by $K \circ \Sigma$.

Figure 3: Product Similarity and Profit Weights (2021)



(a) Figure 3

Figure 4: Combined Network Visualization



(b) Figure 4

6.6 Table 3: Welfare Estimates (2021)

Read: Under common ownership, total surplus is 8.303 trillion dollars; under standard Cournot it is 9.570 trillion dollars.

Detailed interpretation: Common ownership raises aggregate profits but lowers consumer surplus by more than the profit gain.

Economic message: Common ownership is inefficient and redistributive.

TABLE 3: WELFARE ESTIMATES (2021)

	Scenario	*Common Ownership*	*Cournot-Nash*	*Perfect Competition*	*Monopoly*
		(1)	(2)	(3)	(4)
Welfare Statistic	**Variable**	$\mathbf{q}^{\phi}$	$\mathbf{q}^{\Psi}$	$\mathbf{q}^{W}$	$\mathbf{q}^{\Pi}$
Total Surplus (US$ trillions)	$W(\mathbf{q})$	8.303	9.570	11.116	7.629
Aggregate Profits (US$ trillions)	$\Pi(\mathbf{q})$	3.300	2.167	0.000	4.141
Consumer Surplus (US$ trillions)	$CS(\mathbf{q})$	5.003	7.402	11.116	3.488
Total Surplus / Perfect Competition	$\frac{W(\mathbf{q})}{W(\mathbf{q}^W)}$	0.747	0.861	1.000	0.686
Aggregate Profit / Total Surplus	$\frac{\Pi(\mathbf{q})}{W(\mathbf{q})}$	0.397	0.226	0.000	0.543
Consumer Surplus / Total Surplus	$\frac{CS(\mathbf{q})}{W(\mathbf{q})}$	0.603	0.774	1.000	0.457

TABLE NOTES: The table reports the model estimates of aggregate profits, consumer surplus, and total surplus for each of the counterfactual scenarios presented in Section 2.

Table 3: Welfare Estimates (2021)

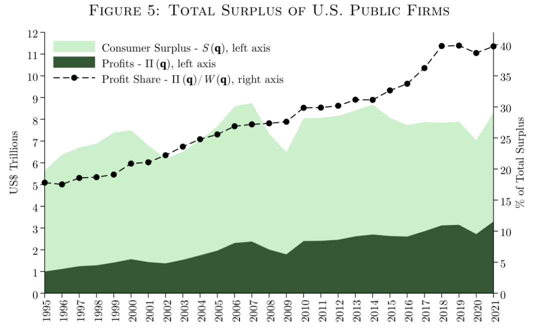
6.7 Figures 5–8: Total Surplus, Profit Share, Deadweight Loss, and Distributional Effects

Read: These figures trace welfare and distributional consequences over 1995–2021.

Detailed interpretation: The common-ownership component grows sharply over time.

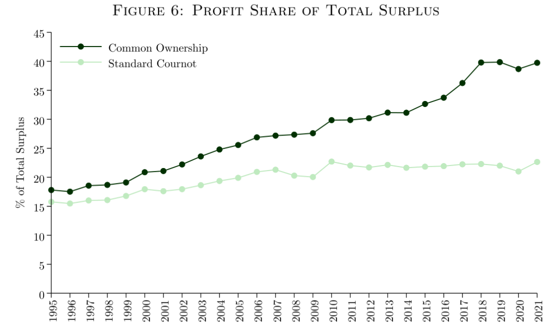
Economic message: Consumer surplus losses and profit gains increase with ownership concentration.

FIGURE 5: TOTAL SURPLUS OF U.S. PUBLIC FIRMS

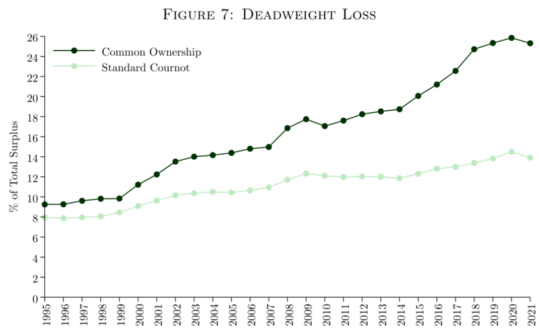


(a) Figure 5

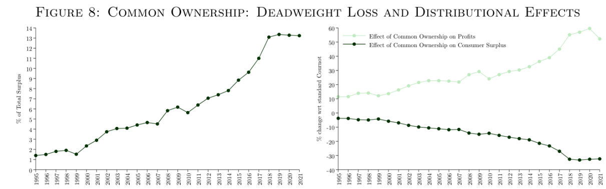
FIGURE 6: PROFIT SHARE OF TOTAL SURPLUS



(b) Figure 6



(a) Figure 7



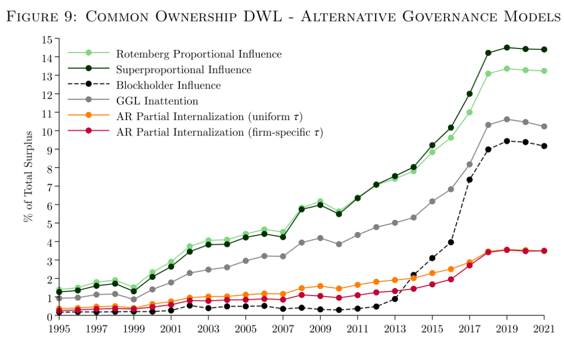
(b) Figure 8

6.8 Figures 9–11: Alternative Governance Models and Extensions

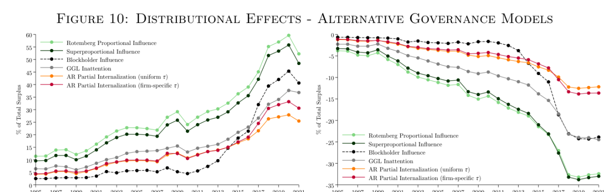
Read: Figures 9 and 10 vary corporate governance assumptions. Figure 11 varies model extensions.

Detailed interpretation: The magnitude changes, but the common-ownership effect remains economically meaningful.

Economic message: The results do not depend on one exact governance specification.



(a) Figure 9



(b) Figure 10

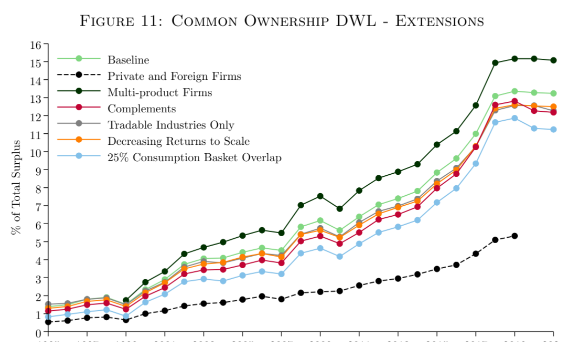


Figure 11: Common Ownership DWL–Extensions

7 Short summary

- The paper quantifies the economy-wide cost of common ownership.
- The key object is the overlap between common ownership and product-market rivalry.
- Common ownership lowers total surplus and shifts surplus from consumers to producers.
- Alternative governance models and extensions preserve the core conclusion.

8 Expanded conclusion

8.1 Core conclusion

The paper shows that common ownership has large welfare and distributional effects in the U.S. public-firm economy.

8.2 Economic intuition

Common owners cause firms to partially internalize competitor profits, softening competition when those competitors are product-market rivals.

8.3 Policy implication

Antitrust analysis should account for ownership networks and their overlap with product-market networks.

8.4 Limitations

The baseline focuses on static unilateral product-market effects under Cournot competition and abstracts from labour-market power, collusion, innovation, and entry.

8.5 Takeaway

Common ownership is a macro-scale competition issue driven by the interaction of two networks.